



**CNP P.H. Robinson to W.A. Parish
345-kV Hardening Project – ERCOT
Independent Review (EIR) Scope**

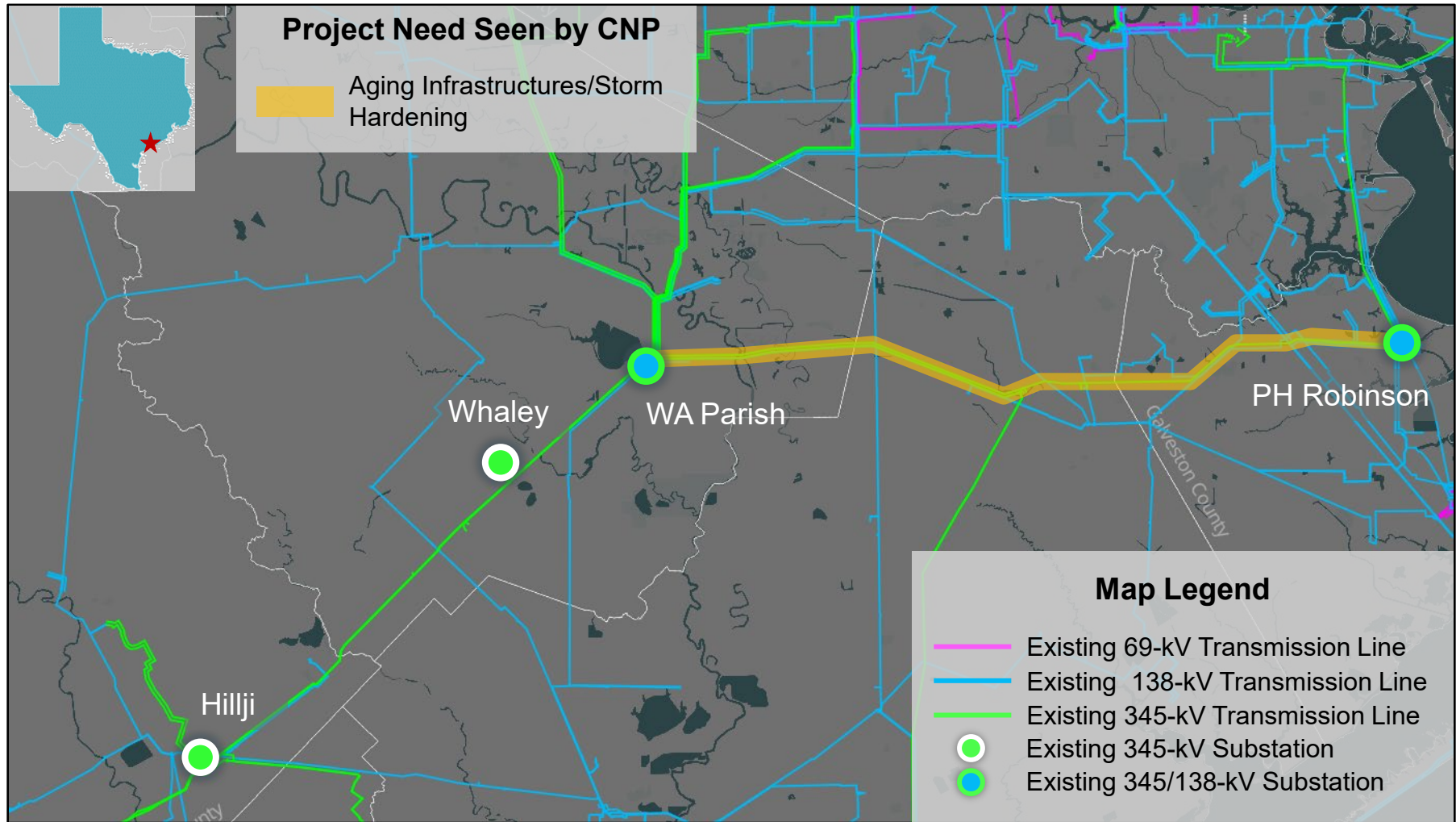
Tanzila Ahmed

RPG Meeting
October 28, 2025

Introduction

- CenterPoint Energy (CNP) submitted the P.H. Robinson to W.A. Parish 345-kV Hardening Project (25RPG028) for Regional Planning Group (RPG) review in August 2025
 - This Tier 1 project is estimated cost to cost \$511.3 million and will not require a Certificate of Convenience and Necessity (CCN)
 - Estimated in-service date (ISD) is December 2028
 - This project is needed to address aging infrastructure and to ensure resilience against hurricane-force winds and extreme cold conditions in the Brazoria, Ft. Bend, and Galveston Counties
- CNP presented project overview and ERCOT presented EIR Scope at the September RPG meeting
 - <https://www.ercot.com/calendar/09252025-RPG-Meeting>
- ERCOT is currently conducting an ERCOT Independent Review (EIR)

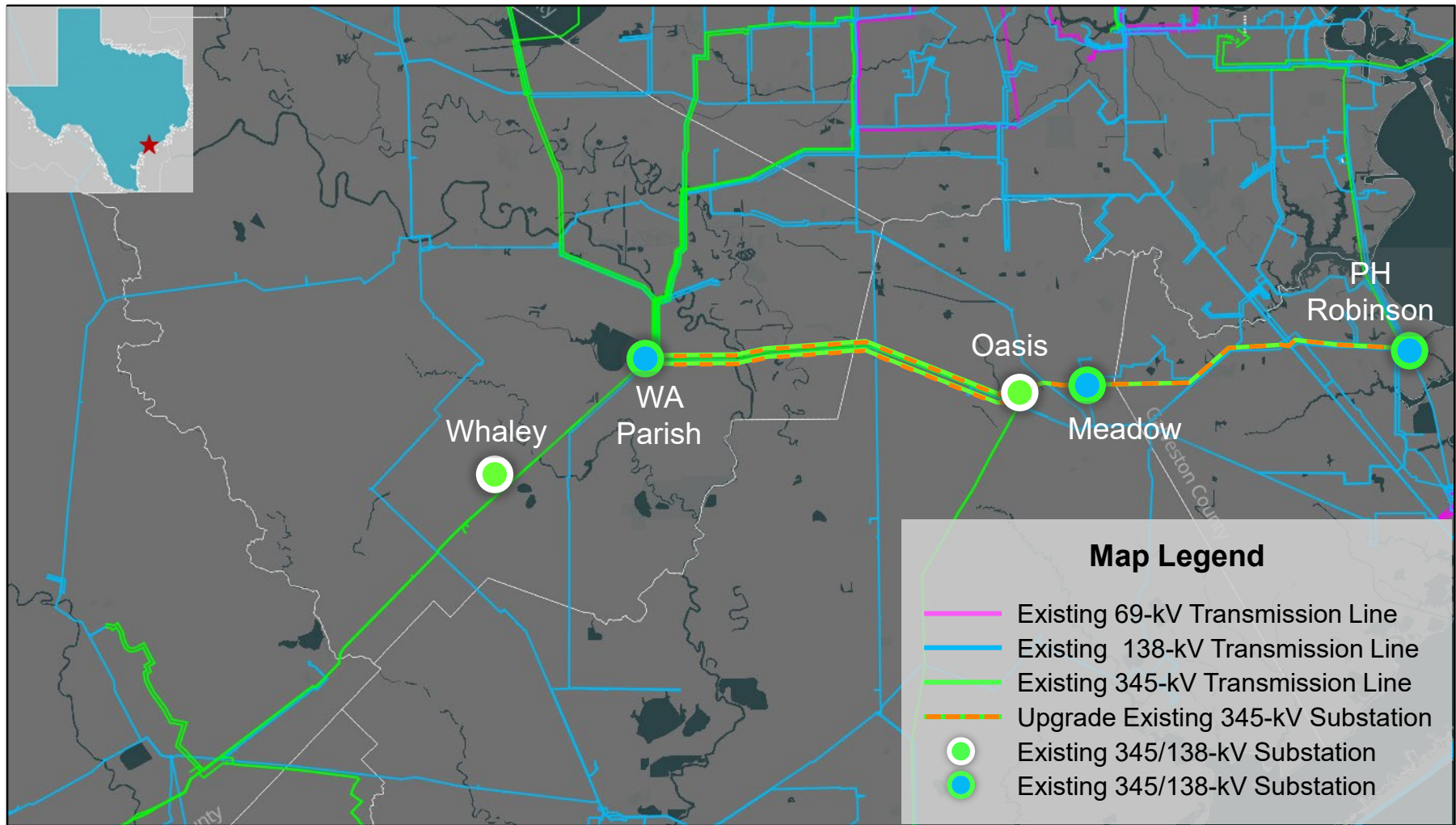
Recap – Map with Project Need Seen by CNP



Recap – Proposed Project by CNP

- Rebuild/reconductor the existing P.H. Robinson to W.A. Parish 345-kV double-circuit transmission line, with a normal and emergency ratings of at least 2,987 MVA, approximately 40.2-mile:
 - Rebuild/reconductor the existing P.H. Robinson to Meadow Ckt.99 345-kV double-circuit transmission line, on double-circuit-capable steel structures with only one circuit in place, approximately 17.4-mile;
 - Ckt.93 will be rebuilt to the North of Ckt.99A within the existing transmission right of way (ROW). Ckt.99 will be rebuilt on double-circuit-capable lattice structures with conductor on both sides of the towers. One side will be Ckt.99, while the other side will be an energized tap;
 - Rebuild the existing Meadow to Oasis Ckt.99 345-kV double-circuit transmission line, on double-circuit-capable steel structures with only one circuit in place, approximately 3.18-mile;
 - Rebuild the existing Oasis to W.A. Parish Ckt.99 and Oasis to W.A. Parish Ckt.18 345-kV double-circuit transmission line on double-circuit-capable steel structures with both circuit in place, approximately 19.6-mile;
 - Upgrade the existing Oasis 345-kV substation to 63 kAIC fault duty breaker rating;
 - Upgrade substation equipment to 5,000A;

Recap – Map with Proposed Project by CNP



Recap – Study Assumptions

- Final 2024 Regional Transmission Planning (RTP) 2029 summer peak load case was used as start case
- Transmission updates
 - New transmission projects, listed in the [Appendix A](#), based on the June 2025 Transmission Project and Information Tracking (TPIT) report along with recently approved RPG projects were added to the study base case
- Generation updates
 - New 6.9(1) generation, listed in [Appendix B](#), were added to the study base case based on the August 2025 Generator Interconnection Status (GIS) report
 - All new generation were dispatched consistent with the 2024 RTP methodology
- Load level in the study area was maintained consistent with the Final RTP case
- The reserve was maintained consistent with the 2024 RTP

Status Update

- Reliability Need Analysis
 - N-1
 - G-1+N-1
 - G-1: Cedar Bayou, South Texas Project (STP), and W.A. Parish generation
 - X-1+N-1
 - X-1: Cedar Bayou, Obrian, P.H. Robinson, and W.A. Parish 345/138-kV transformers
- Option 1 Evaluation
 - Reliability Need Analysis
 - N-1
 - G-1+N-1
 - G-1: Cedar Bayou, South Texas Project (STP), and W.A. Parish generation
 - X-1+N-1
 - X-1: Cedar Bayou, Obrian, P.H. Robinson, and W.A. Parish 345/138-kV transformers

Preliminary Results of Reliability Assessments

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC TPL-001-5.1 and ERCOT Planning Criteria to identify project need and to evaluate the proposed option

	N-1		G-1*+N-1		X-1**+N-1		Unsolved Powerflow
	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	
Base case	None	None	None	None	None	None	None
Option 1	None	None	None	None	None	None	None

* G-1: Cedar Bayou, South Texas Project (STP), and W.A. Parish generations

** X-1: Cedar Bayou, Obrian, and P.H. Robinson 345/138-kV transformers

Next Steps and Tentative Timeline

- Project Evaluation
 - ERCOT may also perform the following studies
 - Planned maintenance outage evaluation
 - Long-Term Load-Serving capability assessment
 - TSP(s) will provide Cost and Feasibility Assessment
- Generation and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
- Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)
- Congestion Analysis
 - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area
- Tentative Timelines
 - Status updates at the future RPG meetings
 - Final recommendation – Q1 2026

Thank you!



Stakeholder comments also welcomed through:

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Appendix A – Transmission Projects Added

- List of recently approved RPG transmission projects added to study base case

RPG/TPIT No	Project Name	Tier	Project ISD	From County
63922	Tiki Island Distribution Upgrades	4	5/31/2026	Galveston
66208	Rebuild League City-Hidden Lakes with bundled 795 ACSS	4	5/31/2026	Galveston
73352A	PH Robinson Area 138kV Upgrades	4	10/31/2026	Galveston
72032	Stewart to West Bay Ckts 48A & 59A Upgrades	4	12/31/2027	Galveston

Appendix B – New Generation Projects Added

- List of new generation, that met the Planning Guide 6.9(1) condition, added to the study base case

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
23INR0029	Cedar Bayou 5 (TEF - Due Diligence)	GAS	12/15/2027	697.0	Chambers
23INR0137	BRP Diran BESS	OTH	1/31/2028	100.6	Wharton
24INR0493	Crowned Heron BESS 2	OTH	3/31/2026	154.2	Fort Bend
24INR0584	Houston IV BESS	OTH	6/3/2026	164.6	Harris
25INR0068	High Chap Solar	SOL	6/26/2028	152.3	Brazoria
25INR0102	Austin Bayou Solar	SOL	6/1/2027	753.4	Brazoria
25INR0103	Elio BESS	OTH	8/5/2027	317.2	Brazoria
25INR0319	Northington Solar	SOL	7/15/2027	129.8	Wharton
25INR0425	Aldrin 345 BESS	OTH	12/1/2027	362.0	Brazoria
26INR0189	Skipjack Energy Storage	OTH	2/28/2028	154.0	Brazoria
26INR0226	First Capitol BESS	OTH	5/1/2027	256.2	Brazoria
26INR0250	Duffy BESS	OTH	9/15/2026	235.9	Matagorda
26INR0333	VERTUS ENERGY STORAGE	OTH	2/1/2026	200.0	Galveston
26INR0405	Buffalo Creek BESS	OTH	6/1/2026	250.0	Fort Bend
27INR0224	Leopard BESS	OTH	7/9/2028	264.2	Victoria